

Explainable Retrieval-Augmented Generation for Cybersecurity Forensics: A Mini Research Report

A Study on Grounded, Transparent, and Auditable AI Systems for Digital Evidence
Analysis

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Abstract

Large Language Models (LLMs) have demonstrated remarkable capabilities in natural language understanding and generation, yet their tendency to hallucinate factual information limits their trustworthiness in high-stakes domains such as digital forensics and cybersecurity investigation. Retrieval-Augmented Generation (RAG) addresses this limitation by grounding model outputs in externally retrieved evidence, thereby improving factual accuracy and traceability. However, RAG systems alone do not guarantee interpretability: investigators and auditors require insight into why a particular document was retrieved and how it influenced the final answer. This report examines the integration of Explainable Artificial Intelligence (XAI) principles into RAG pipelines specifically designed for cybersecurity forensic workflows. We describe a reference architecture combining semantic embedding retrieval, vector similarity search, and citation-aware generation, and we discuss evaluation strategies, limitations, and opportunities for future research. The report is intended as a practical reference for building small-scale, explainable, retrieval-grounded assistants suitable for academic prototypes and dissertation-level experimentation.

Keywords: Explainable AI, Retrieval-Augmented Generation, Digital Forensics, Vector Databases, Large Language Models, Cybersecurity, Interpretability

1. Introduction

1.1 Background

The rapid adoption of Large Language Models across enterprise and research settings has introduced a new class of intelligent assistants capable of summarizing, reasoning about, and generating natural language responses to complex queries. In cybersecurity, these systems are increasingly proposed for tasks such as incident report summarization, log analysis, malware behavior description, and forensic timeline reconstruction. However, LLMs are fundamentally statistical text predictors trained on fixed corpora, which makes them prone to two critical failure modes: hallucination, where the model generates plausible but false statements, and knowledge staleness, where the model lacks awareness of recent or domain-specific evidence.

Retrieval-Augmented Generation was introduced as a mechanism to mitigate these failure modes by coupling a parametric language model with a non-parametric retrieval component. Instead of relying solely on knowledge encoded during training, a RAG system first retrieves relevant passages from an external knowledge base and then conditions the generation process on this retrieved context. This architecture has become a de facto standard for building domain-specific assistants, particularly in fields where factual precision and traceability are essential, such as law, medicine, and digital forensics.

1.2 Motivation

Digital forensics investigators are often required to justify their conclusions in legal proceedings. A forensic assistant that produces an answer without indicating which piece of evidence supports that answer is of limited practical value, regardless of its accuracy. This creates a strong motivation for combining RAG with explainability techniques: not only should the system retrieve the correct evidence, but it should also present that evidence transparently, allow the investigator to trace each claim back to its source document, and avoid presenting information that is not directly supported by retrieved material.

This report is motivated by the practical need to design small, auditable, and reproducible RAG prototypes that can later be extended into larger dissertation-scale systems. The focus is deliberately kept narrow: rather than proposing a completely novel architecture, the report consolidates established techniques into a coherent, explainable pipeline suitable for academic experimentation.

1.3 Objectives

- To describe a reference architecture for explainable, retrieval-grounded question answering in forensic contexts.
- To identify the key components required for transparent citation and source attribution.
- To discuss evaluation metrics suitable for measuring both retrieval quality and generation faithfulness.

- To highlight limitations and open research questions relevant to dissertation-level extension.

2. Related Work

2.1 Retrieval-Augmented Generation

The foundational RAG framework combines a dense passage retriever with a sequence-to-sequence generator, enabling the model to condition its output on documents retrieved from an external corpus at inference time. Subsequent work has extended this idea in multiple directions, including iterative retrieval, where the model issues multiple retrieval queries during generation, and self-reflective retrieval, where the model evaluates whether retrieved evidence is sufficient before producing an answer. These extensions are particularly relevant to forensic use cases, where a single retrieval pass may not surface all relevant evidence scattered across multiple documents or log files.

2.2 Explainable Artificial Intelligence

Explainable AI research has traditionally focused on interpreting the internal decision process of classification and regression models, using techniques such as feature attribution, saliency mapping, and surrogate modeling. In the context of generative language models, explainability takes a different form: rather than explaining internal neuron activations, the emphasis shifts toward explaining which external evidence informed a generated statement. This form of explainability is sometimes referred to as attribution-based or citation-based explainability, and it aligns naturally with the retrieval component of RAG systems.

2.3 AI in Digital Forensics

Prior work in digital forensics has explored the use of machine learning for malware classification, network intrusion detection, and automated evidence triage. More recently, researchers have begun investigating the use of language models to assist investigators in summarizing large volumes of unstructured evidence, such as chat logs, emails, and system event records. A recurring concern in this literature is the risk of introducing unverified or fabricated claims into a forensic report, which underscores the importance of grounding generated text in verifiable source material rather than relying on model memory alone.

2.4 Vector Databases and Semantic Search

Modern RAG systems rely heavily on vector databases such as Chroma, FAISS, Pinecone, and Weaviate to store dense embedding representations of documents and to perform approximate nearest neighbor search at query time. These systems convert text into fixed-length numerical vectors using pretrained embedding models, and retrieve the most semantically similar vectors to a given query using distance metrics such as cosine similarity or Euclidean distance. The choice of embedding model, chunking strategy, and similarity metric each has a measurable impact on retrieval quality, which in turn affects the factual accuracy of the final generated answer.

3. Proposed System Architecture

3.1 Overview

The proposed reference architecture consists of five sequential stages: document ingestion, chunking, embedding generation, vector storage and retrieval, and grounded answer generation with citation. Each stage is designed to be independently testable, which is particularly useful in an academic prototyping context where components are frequently swapped or upgraded during experimentation.

3.2 Document Ingestion

Forensic evidence is typically heterogeneous, spanning PDF reports, plain text logs, and structured metadata. The ingestion stage is responsible for extracting raw text from these sources while preserving provenance information such as file name, page number, and timestamp. Preserving this metadata is essential, since it forms the basis for later citation and traceability in the generated output.

3.3 Chunking Strategy

Extracted text is divided into overlapping chunks prior to embedding. Chunk size and overlap represent a trade-off between retrieval precision and contextual completeness. Smaller chunks tend to produce more precise retrieval but risk losing surrounding context, while larger chunks preserve context but may dilute semantic focus and increase the likelihood of retrieving partially irrelevant material. In practice, chunk sizes between approximately 500 and 1000 characters, with an overlap of 10 to 20 percent, provide a reasonable balance for typical forensic documents.

3.4 Embedding and Vector Storage

Each chunk is transformed into a dense vector representation using a pretrained sentence embedding model. These vectors are stored in a persistent vector database alongside their associated metadata. At query time, the user's question is embedded using the same model, and the vector database performs a similarity search to identify the most relevant chunks. Consistency between the embedding model used during ingestion and the model used during querying is critical, since mismatched embedding spaces produce meaningless similarity scores.

3.5 Grounded Generation with Citation

Retrieved chunks, together with their source metadata, are inserted into a structured prompt that instructs the language model to answer strictly based on the provided context. The prompt explicitly requires the model to decline answering when the retrieved context does not contain sufficient information, and to include inline citations referencing the specific source document and page number for each factual claim. This constraint transforms the language model from an open-ended generator into a constrained, evidence-bound reasoning component, which is essential for forensic applicability.

4. Explainability Layer

4.1 Retrieval Transparency

Retrieval transparency refers to the ability to inspect which documents were retrieved for a given query and why. In the proposed architecture, every retrieved chunk is logged along with its similarity score, source file, and page number. This log can be surfaced directly to the investigator, allowing manual verification of retrieval relevance independent of the final generated answer.

4.2 Attribution and Citation Consistency

A key explainability requirement is attribution consistency, meaning that every claim in the generated answer should correspond to content actually present in the cited source. Automated verification of this property can be performed by checking whether the semantic content of each cited chunk supports the corresponding sentence in the generated answer. Discrepancies between cited and actual content indicate either retrieval failure or generation drift, both of which should be flagged for manual review.

4.3 Confidence and Refusal Behavior

An explainable forensic assistant should exhibit calibrated refusal behavior, explicitly stating when available evidence is insufficient to answer a question, rather than generating a plausible but unsupported response. This behavior can be enforced through prompt engineering, and can be further validated through targeted test queries that intentionally fall outside the scope of the indexed document collection.

5. Evaluation Methodology

5.1 Retrieval Quality Metrics

Retrieval quality can be evaluated using standard information retrieval metrics such as precision at k, recall at k, and mean reciprocal rank, computed against a manually annotated set of question-document relevance pairs. These metrics quantify how effectively the system surfaces relevant evidence before any generation takes place.

5.2 Generation Faithfulness

Faithfulness measures the degree to which generated statements are supported by retrieved context. This can be assessed manually through expert review, or automatically using natural language inference models that classify each generated sentence as entailed, contradicted, or unsupported by the corresponding cited chunk.

5.3 End-to-End Task Accuracy

Beyond retrieval and faithfulness, end-to-end accuracy evaluates whether the final answer correctly addresses the user's question in a manner useful to a forensic investigator. This typically requires a curated evaluation set of realistic forensic queries paired with reference answers derived from ground-truth case documentation.

Table 1. Summary of evaluation dimensions relevant to explainable RAG systems in forensic applications.

Dimension	Metric	Purpose
Retrieval Quality	Precision@k, Recall@k, MRR	Measures relevance of retrieved evidence
Faithfulness	Entailment classification	Detects unsupported or hallucinated claims
Attribution	Citation-content match rate	Verifies citation correctness
Refusal Calibration	False-answer rate on out-of-scope queries	Ensures safe refusal behavior
Task Accuracy	Human expert scoring	Assesses practical investigative usefulness

6. Illustrative Case Scenario

Consider a simplified scenario in which an investigator has collected a set of documents related to a suspected data exfiltration incident, including a network log summary, an incident response report, and an internal security policy document. An explainable RAG assistant configured over this document set could be asked: "What evidence suggests unauthorized data transfer occurred?" The system would first retrieve chunks from the network log summary describing unusual outbound traffic patterns, then generate a response that explicitly cites the specific log entry and corresponding page number. If the investigator subsequently asks a question unrelated to the indexed documents, such as a query about an unrelated case, the system should explicitly state that it lacks sufficient information rather than fabricating a plausible-sounding answer.

This scenario illustrates the practical value of combining retrieval grounding with explicit refusal behavior: the system remains useful for evidence synthesis while avoiding the introduction of unverified claims into a forensic narrative, which could otherwise compromise the admissibility or credibility of an investigative report.

6.1 Sample Interaction Trace

Query: What evidence suggests unauthorized data transfer occurred?

Retrieved Sources: network_log_summary.pdf (Page 3), incident_response_report.pdf (Page 5)

Generated Answer: The network log summary indicates an unusual spike in outbound traffic to an external IP address outside normal business hours [Page 3]. The incident response report corroborates this observation, noting that the affected host had not previously communicated with external endpoints in the preceding thirty days [Page 5].

7. Limitations

- Retrieval quality is highly dependent on chunking strategy and embedding model choice, both of which may require domain-specific tuning for forensic terminology.
- Citation-based explainability does not guarantee that the underlying language model has not incorporated implicit biases from its pretraining data, even when constrained to retrieved context.
- The evaluation of faithfulness through automated entailment models remains imperfect and should be complemented with human expert review in high-stakes forensic contexts.
- Small-scale prototypes described in this report have not been tested against adversarial or intentionally misleading document collections, which represents an important direction for future robustness evaluation.
- Latency introduced by the retrieval and generation pipeline may be non-trivial for very large forensic document collections, requiring further optimization for production deployment.

8. Future Work

Several directions extend naturally from this reference architecture. First, integrating hybrid retrieval, combining sparse keyword-based search with dense semantic search, could improve recall for queries containing specific technical identifiers such as IP addresses or file hashes that dense embeddings may not represent precisely. Second, incorporating a reranking stage after initial retrieval could improve precision by applying a more computationally expensive but more accurate relevance model to a smaller candidate set. Third, extending the explainability layer to include automated citation verification, using natural language inference to flag unsupported claims before presenting an answer to the investigator, would strengthen the reliability guarantees of the system. Finally, evaluating the architecture on real-world, ethically sourced forensic datasets, with appropriate privacy safeguards, would provide stronger empirical validation than the illustrative scenario discussed in this report.

9. Conclusion

This report has outlined a reference architecture for explainable, retrieval-grounded question answering in cybersecurity forensic contexts. By combining semantic retrieval, persistent vector storage, and citation-constrained generation, it is possible to construct small-scale prototypes that are both practically useful and transparent in their reasoning. The proposed evaluation methodology, spanning retrieval quality, generation faithfulness, attribution consistency, and refusal calibration, provides a structured basis for assessing such systems during dissertation-level research. While several limitations remain, particularly around adversarial robustness and large-scale deployment, the architecture described here offers a practical starting point for further academic investigation into explainable AI for digital forensics.

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